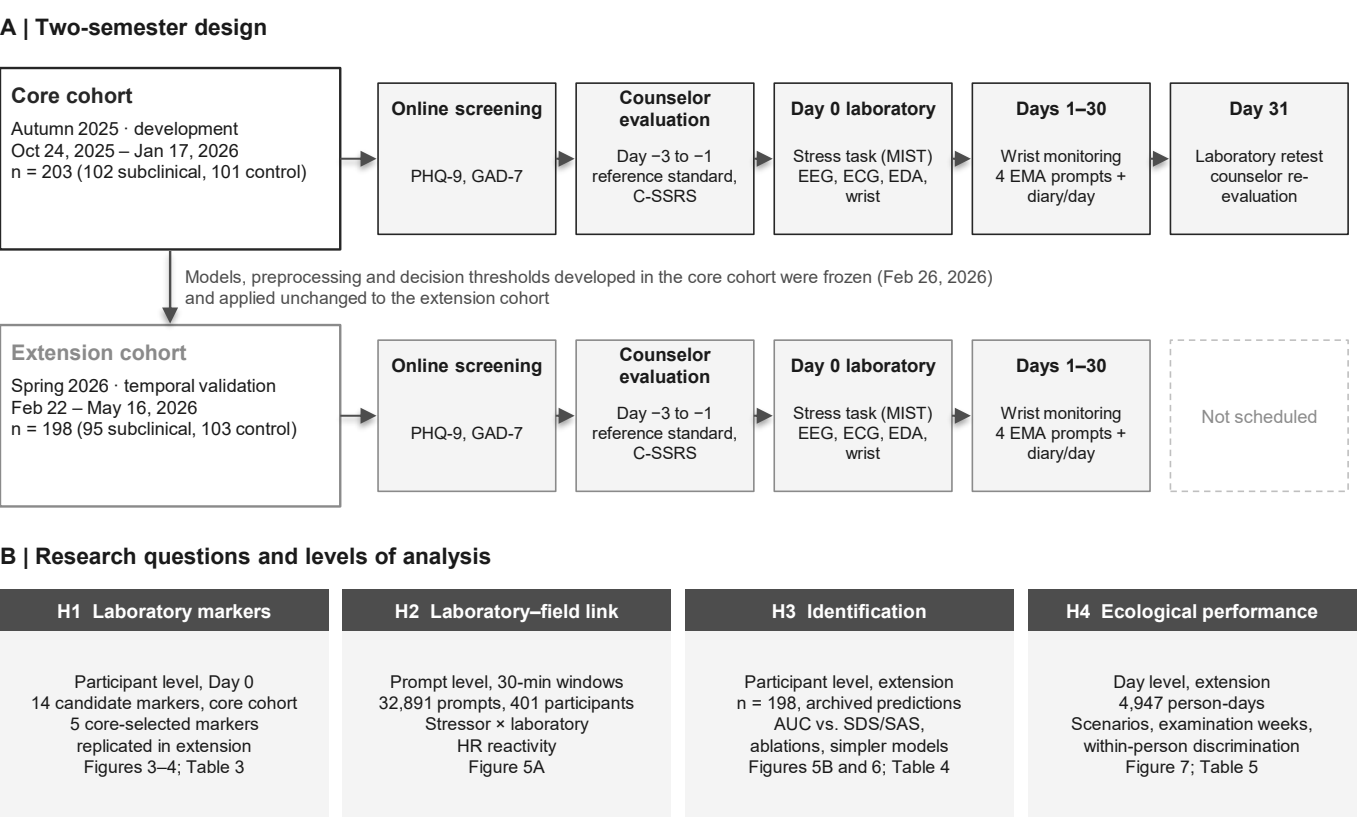

A | Two-semester design
Core cohort
Autumn 2025 · development
Oct 24, 2025 – Jan 17, 2026
n = 203 (102 subclinical, 101 control)
Online screening
PHQ-9, GAD-7
Counselor evaluation
Day −3 to −1 reference standard, C-SSRS
Day 0 laboratory
Stress task (MIST) EEG, ECG, EDA, wrist
Days 1–30
Wrist monitoring 4 EMA prompts + diary/day
Day 31
Laboratory retest counselor re-evaluation
Models, preprocessing and decision thresholds developed in the core cohort were frozen (Feb 26, 2026) and applied unchanged to the extension cohort
Extension cohort
Spring 2026 · temporal validation
Feb 22 – May 16, 2026
n = 198 (95 subclinical, 103 control)
Online screening
PHQ-9, GAD-7
Counselor evaluation
Day −3 to −1 reference standard, C-SSRS
Day 0 laboratory
Stress task (MIST) EEG, ECG, EDA, wrist
Days 1–30
Wrist monitoring 4 EMA prompts + diary/day
Not scheduled
B | Research questions and levels of analysis
H1 Laboratory markers
Participant level, Day 0
14 candidate markers, core cohort
5 core-selected markers replicated in extension
Figures 3–4; Table 3
H2 Laboratory–field link
Prompt level, 30-min windows
32,891 prompts, 401 participants
Stressor × laboratory HR reactivity
Figure 5A
H3 Identification
Participant level, extension
n = 198, archived predictions
AUC vs. SDS/SAS, ablations, simpler models
Figures 5B and 6; Table 4
H4 Ecological performance
Day level, extension
4,947 person-days
Scenarios, examination weeks, within-person discrimination
Figure 7; Table 5